

ROSARIO, Argentina

WMO: 874800

Lat: 32.9081S

Lon: 60.7822W

Elev: 25

StdP: 101.03

Time Zone: -3.00 (W03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	-0.4	1.1	-4.9	2.5	6.9	-3.0	2.9	7.0	10.9	13.9	9.9	14.5	1.3	180	0.427

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.5	34.5	23.3	33.1	22.9	31.9	22.5	25.8	31.3	24.9	30.1	24.1	29.2	4.8	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.2	19.2	28.7	23.2	18.1	27.8	22.4	17.2	27.0	80.1	31.1	76.1	30.1	72.7	29.2	30.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.1	8.7	7.6	-3.6	37.5	1.4	1.3	-4.6	38.5	-5.4	39.2	-6.2	40.0	-7.2	41.0		
(5)			-4.2	28.1	1.3	1.3	-5.2	29.0	-5.9	29.8	-6.7	30.5	-7.6	31.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	17.9	24.8	23.4	21.5	18.0	14.7	11.6	10.9	12.6	15.1	18.2	21.2	23.6	(6)
(7)		DBStd	5.92	2.79	3.07	3.24	3.68	3.80	3.82	4.08	4.29	3.91	3.42	3.12	3.02	(7)
(8)		HDD10.0	87	0	0	0	1	4	23	36	20	4	0	0	0	(8)
(9)		HDD18.3	980	1	2	9	49	127	206	233	187	112	44	9	2	(9)
(10)		CDD10.0	2982	458	377	357	240	148	71	64	100	156	254	335	422	(10)
(11)		CDD18.3	834	201	145	107	39	13	4	4	8	14	40	94	165	(11)
(12)		CDH23.3	7969	2114	1371	909	279	64	10	21	74	149	368	919	1691	(12)
(13)	CDH26.7	3013	927	533	295	63	8	1	2	17	39	102	321	705	(13)	
(14)	Wind	WSAvg	3.4	3.2	3.0	2.9	3.0	2.8	3.0	3.3	3.6	4.0	4.1	3.9	3.5	(14)
(15)	Precipitation	PrecAvg	1007	109	116	155	96	52	34	31	36	57	107	105	122	(15)
(16)		PrecMax	1625	238	394	509	265	272	134	112	151	238	331	234	333	(16)
(17)		PrecMin	667	9	5	24	1	0	0	0	0	1	6	17	30	(17)
(18)		PrecStd	211	60	88	99	62	48	33	27	38	48	65	56	69	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.0	35.1	33.7	31.1	27.9	24.9	26.0	29.0	30.9	32.1	34.1	36.1	(19)
MCWB			23.8	24.1	24.1	23.1	20.9	20.1	18.7	20.4	20.6	20.8	21.6	23.5	(20)	
2%		DB	34.8	33.2	31.5	28.3	25.0	21.8	22.4	25.2	27.0	29.3	32.0	34.0	(21)	
		MCWB	23.4	23.9	22.9	20.9	19.6	18.1	17.2	18.1	18.7	20.0	21.0	22.7	(22)	
5%		DB	33.0	31.9	29.9	26.4	22.9	19.8	19.9	22.6	24.8	27.1	30.1	32.2	(23)	
		MCWB	23.0	23.6	22.3	20.1	18.5	16.4	15.2	16.3	16.9	19.0	20.2	22.1	(24)	
10%		DB	31.4	30.0	28.1	24.7	21.0	18.0	17.8	20.1	22.5	25.1	28.2	30.8	(25)	
		MCWB	22.5	22.7	21.3	19.3	17.2	14.6	13.9	14.7	15.7	17.9	19.4	21.6	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.8	27.0	26.4	24.5	22.1	20.8	19.9	21.2	22.0	23.7	24.1	26.6	(27)
MCDB			32.8	32.8	31.4	28.7	25.9	23.7	23.9	26.9	28.7	28.9	30.7	32.2	(28)	
2%		WB	25.5	25.7	24.7	22.9	20.5	19.0	18.0	19.4	20.0	21.7	22.5	24.9	(29)	
		MCDB	31.5	30.7	28.9	26.5	23.6	20.9	21.6	24.2	25.4	26.5	28.8	30.3	(30)	
5%		WB	24.6	24.6	23.3	21.6	19.3	17.4	16.5	17.7	18.4	20.4	21.5	23.8	(31)	
		MCDB	30.4	29.4	27.6	24.8	22.2	19.2	19.3	21.4	22.5	25.3	27.5	29.3	(32)	
10%		WB	23.8	23.6	22.3	20.4	18.0	15.5	15.0	15.7	16.9	19.2	20.6	22.7	(33)	
		MCDB	29.1	28.3	26.5	23.2	20.4	17.1	16.9	18.8	21.1	23.6	26.2	28.1	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	11.5	10.7	10.9	10.6	9.9	10.2	10.9	11.9	11.2	11.8	11.7	(35)	
MCDBR			13.7	12.7	12.9	12.9	11.5	11.0	12.9	14.5	14.8	14.2	14.5	14.2	(36)	
MCWBR			5.4	5.3	5.7	6.5	6.5	7.1	7.7	8.0	7.4	6.8	6.2	5.6	(37)	
5% WB		MCDBR	11.2	10.4	10.5	9.7	9.6	8.3	10.5	11.6	11.7	11.6	12.3	11.3	(38)	
		MCWBR	5.3	5.3	5.4	5.8	5.8	6.3	7.0	7.0	6.8	6.6	6.1	5.6	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.406	0.397	0.373	0.378	0.358	0.354	0.355	0.418	0.475	0.423	0.399	0.399	(40)	
taud		2.354	2.393	2.458	2.414	2.427	2.426	2.397	2.176	2.019	2.239	2.334	2.360	(41)		
Ebn at Noon		933	921	908	843	805	781	798	784	792	885	933	944	(42)		
Edh at Noon		132	123	108	101	88	83	90	125	164	142	134	132	(43)		
(44)	All-Sky Solar Radiation	RadAvg	7.16	6.16	5.25	3.82	2.81	2.40	2.64	3.53	4.68	5.65	6.84	7.27	(44)	
(45)		RadStd	0.40	0.49	0.47	0.47	0.25	0.28	0.28	0.34	0.39	0.49	0.41	0.44	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	+0.18	+0.59	+0.63	N/A	N/A	N/A	N/A
(47)	Regional (4 neighbors)	N/A	N/A	N/A	+0.35	+0.50	+0.54	N/A	N/A	N/A	N/A

Nomenclature: See separate page